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(*Algorithm:LM-STLS (Levenberg-Marquardt Sequential Thresholded Least Squares)
Reference to cite when using this code:[1] A noise-corrected data-
driven approach for nonlinear dynamical system modeling in frequency domain,2026.
Code availability:This program is open for academic research
only.Unauthorized commercial redistribution is prohibited.
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=====*)

(*Import self-defined package for nonlinear sparse identification*)
Import["NCSINDyPackage.m", "Package"]

(* =====Basic System Configuration=====*)
(*n:nonlinear polynomial order;order:Fourier series expansion order*)
n = 7;
order = 5;
opts = {AccuracyGoal → 12, PrecisionGoal → 12, MaxStepFraction → 1/1000 };
(*Governing differential equation of nonlinear oscillator*)
Eq01 = {m x''[t] + c x'[t] +  $\sum_{j1=1}^n k_{j1} x[t]^{j1} - F \cos[\Omega t] == 0$ };

(*Initial conditions of dynamic system*)
InitCond01 = {x[0] == 0, x'[0] == 0};
(*True physical parameter values for simulation*)
Par01 = Join[{m → 1, c → 0.05, F → 1},
Thread[Table[kj1, {j1, 1, n}] → {1, 0.4, 0.5, 0, 0.2, 0, 0}]];
(*List of undetermined physical coefficients to be identified*)
Var = Join[{m, c}, Table[kj1, {j1, 1, n}]];

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(* =====
Fourier Series Model Definition=====*)
(*Displacement,velocity,acceleration Fourier series model*)
(*DispModel:displacement Fourier approximation*)
DispModel =  $A_{s,0} + \sum_{j1=1}^{order} (A_{s,j1} \cos[j1 \Omega t] + B_{s,j1} \sin[j1 \Omega t])$ ;

(*Coefficient list for displacement Fourier series*)
DispCoeff = Join[{As,0}, Flatten[Table[{As,j1, Bs,j1}, {j1, order}]]];
(*VeloModel:velocity Fourier approximation*)
VeloModel =  $A_{v,0} + \sum_{j1=1}^{order} (A_{v,j1} \cos[j1 \Omega t] + B_{v,j1} \sin[j1 \Omega t])$ ;

VeloCoeff = Join[{Av,0}, Flatten[Table[{Av,j1, Bv,j1}, {j1, order}]]];
(*AcceModel:acceleration Fourier approximation*)
AcceModel =  $A_{a,0} + \sum_{j1=1}^{order} (A_{a,j1} \cos[j1 \Omega t] + B_{a,j1} \sin[j1 \Omega t])$ ;

AcceCoeff = Join[{Aa,0}, Flatten[Table[{Aa,j1, Ba,j1}, {j1, order}]]];
(*Harmonic orders used for noise correction*)
OrderCorrect = {4, 5};
(*Coefficient set of noise correction terms*)
CorrectModel = Sum[ $\delta A_{a,j1} \cos[j1 \Omega t] + \delta B_{a,j1} \sin[j1 \Omega t]$ , {j1, OrderCorrect}];
(*Coefficient set of noise correction terms*)
CorrectCoeff = DeleteCases[Flatten[Table[{ $\delta A_{a,j1}$ ,  $\delta B_{a,j1}$ }, {j1, OrderCorrect}]]],  $\delta B_{a,0}$ ];
(*Sub-function:construct correction coefficients for each excitation frequency*)
CorrectCoeffj[jj_] :=
  Join[{ $\delta A_{a,0}[jj]$ }, Flatten[Table[{ $\delta A_{a,j1}[jj]$ ,  $\delta B_{a,j1}[jj]$ }, {j1, order}]]] /.
  x1_x2_,x3_[x4_] /; MemberQ[Complement[Range[0, order], OrderCorrect], x3] → 0;
(*Extract variable symbols of noise correction coefficients*)
CorrectCoeffjVar[jj_] :=
  DeleteCases[Flatten[Table[{ $\delta A_{a,j1}[jj]$ ,  $\delta B_{a,j1}[jj]$ }, {j1, OrderCorrect}]]],  $\delta B_{a,0}[jj]$ ];

(*Nonlinear stiffness model library constructed from displacement harmonics*)
StiffnessModel = ModelLibFun[{DispCoeff}, n, order];
(*Excitation frequency list for sweeping analysis*)
OmegaList = Range[0.5, 4, 0.5];
(*Sampling time sequence*)TimeData = Range[800, 1000, 0.01];
(*Signal-to-Noise Ratio*)
SNR = 10;

(*Load preset noise data from external file for reproducibility*)
Noise = Transpose[Import["C:/Users/Administrator/Desktop/Noise42.xls"]][[1]];

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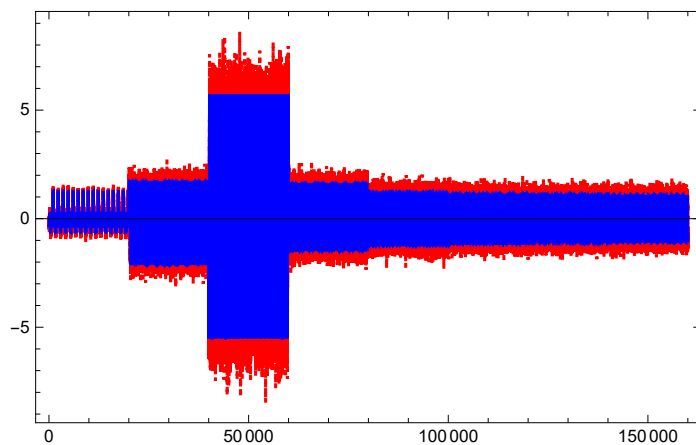
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(* =====Generate Dynamic Response Data=====*)
For[j2 = 1, j2 ≤ Length[OmegaList], j2++,
  (*Solve ODE to obtain true displacement response*)
  Sol01[j2] = NDSolveValue[
    Join[Eq01, InitCond01] /. Par01 /. Ω → OmegaList[[j2]], x, {t, 0, 1000}, opts];
  (*True acceleration time series data*)
  AcceData[j2] =
    Array[{TimeData[[#]], Sol01[j2]''[TimeData[[#]]]} &, Length[TimeData]];
  AcceSigTrue[j2] = Array[Sol01[j2]''[TimeData[[#]]] &, Length[TimeData]];
  (*Add preset noise to obtain measured noisy acceleration*)
  (*AcceSigNois[j2]=AddNoiseFun[AcceSigTrue[j2],0,10];*)
  AcceSigNois[j2] = Noise[[j2]];
  (*AcceSigNois[j2]=Noise[[j2]];*)
  AcceDataMeas[j2] = Array[
    {TimeData[[#]], AcceSigTrue[j2] [[#]] + AcceSigNois[j2] [[#]]} &, Length[TimeData]];
  (* =====
  Fourier Coefficient Identification via Fitting=====*)
  (*Fourier coefficients fitted from noisy measured data*)
  Ameas[x''] [j2] = AcceCoeff /.
    FindFit[AcceDataMeas[j2], AcceModel /. Par01 /. Ω → OmegaList[[j2]], AcceCoeff, t];
  (*Fourier coefficients fitted from noise-free true data*)
  Atrue[x''] [j2] = AcceCoeff /.
    FindFit[AcceData[j2], AcceModel /. Par01 /. Ω → OmegaList[[j2]], AcceCoeff, t];
  (*Fourier coefficients of pure noise component*)
  Anois[x''] [j2] = CorrectCoeff /.
    FindFit[Array[{TimeData[[#]], AcceSigNois[j2] [[#]]} &, Length[TimeData]],
      CorrectModel /. Par01 /. Ω → OmegaList[[j2]], CorrectCoeff, t];
  (*Noise correction for acceleration harmonic coefficients*)
  A[x''] [j2] =
    ReplacePart[Ameas[x''] [j2] - (CorrectCoeffj[j2] /. Ω → OmegaList[[j2]]), 1 → 0];
  (* =====
  Full-State Reconstruction via SVA Transform=====*)
  (*Reconstruct velocity and acceleration from corrected displacement harmonics*)
  A[x] [j2] = ReplacePart[
    SVATransform[A[x''] [j2], "rad/s", OmegaList[[j2]], "a2s"], 1 → δAs,0[j2]];
  A[x'] [j2] = ReplacePart[SVATransform[A[x''] [j2],
    "rad/s", OmegaList[[j2]], "a2v"], 1 → 0];
  (*Excitation force harmonic coefficients*)
  A[f] [j2] = Join[{0, F}, Table[0, 2 order - 1]] /. Par01;
  (*Construct nonlinear library matrix in frequency domain*)
  ModelLibrary[j2] = StiffnessModel /. Thread[DispCoeff → A[x] [j2]];
  (*Assemble Jacobian-like matrix and residual vector for nonlinear least squares*)
  AA1[j2] = Join[{A[x''] [j2]}T, {A[x'] [j2]}T, ModelLibrary[j2], 2].{Var}T // Chop;
  RR1[j2] = Flatten[AA1[j2] - {A[f] [j2]}T];
  Print[j2/Length[OmegaList] // N];
]

```

0.125
 0.25
 0.375
 0.5
 0.625
 0.75
 0.875
 1.

```
(* =====
  Prepare Training Data for Sparse Regression=====*)
(*Collect noisy and noise-free training time-series data*)
TrainDataAcceMeas = Table[AcceDataMeas[j1], {j1, 1, Length[OmegaList]}];
TrainDataAcceTrue = Table[AcceData[j1], {j1, 1, Length[OmegaList]}];
(*Plot noisy and true acceleration responses*)
Fig2aNoise =
  ListLinePlot[Flatten[Table[AcceDataMeas[j1][[All, 2]], {j1, 1, Length[OmegaList]}]],
    PlotStyle -> {Red, Dotted}, Frame -> True, PlotRange -> All];
Fig2aTrue = ListLinePlot[Flatten[Table[AcceData[j1][[All, 2]],
  {j1, 1, Length[OmegaList]}]], PlotStyle -> {Blue}, Frame -> True, PlotRange -> All];
Fig2a = Show[Fig2aNoise, Fig2aTrue, PlotRange -> All]
```



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(* =====
  LM-STLS Algorithm Hyperparameters=====*)
ConvTol = 10-6; maxLMIter = 100;
Lambda = 0.06; maxSparseIter = 10;
(*Combine residual across all excitation frequencies*)
Residual = Join@@ Table[RR1[j2], {j2, 1, Length[OmegaList]}];
```

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(*Case1: the streaming corruption is not corrected,
the damping is not protected, Lambda is too large*)
(*Combine residual across all excitation frequencies*)
Residual01 = Residual /. Thread[Table[ $\delta A_{s,0}[j2]$ , {j2, 1, Length[OmegaList]}]  $\rightarrow$  0];
(*Merge physical parameters and noise
correction coefficients into full variable set*)
Variable01 = Join[Var,
  Table[Join[CorrectCoeffjVar[j2]], {j2, 1, Length[OmegaList]}] // Flatten];
(*Compute Jacobian matrix of residual with respect to unknown variables*)
JacobiFun01 = D[Residual01, {Variable01, 1}];
(*Initial guess of unknown parameter vector*)
InitGuess01 = ReplacePart[ReplacePart[0.2 * Variable01 /. Par01,
  Thread[Range[Length[Var] + 1, Length[Variable01]]  $\rightarrow$  0]], Thread[{6, 8, 9}  $\rightarrow$  1]];
(*Indices of physical terms forced to be retained during thresholding*)
InitbigIndices01 = Join[Range[Length[Var] + 1, Length[Variable01]]];

(*Execute proposed LM-STLS nonlinear sparse regression*)
Xi01 = NonLinearSparsifyDynamics[JacobiFun01, Residual01, Variable01,
  InitGuess01, ConvTol, maxLMIter, Lambda, maxSparseIter, InitbigIndices01];

{1, 38}
{2, 37}
{3, 37}
{4, 37}
{5, 37}
{6, 37}
{7, 37}
{8, 37}
{9, 37}
{10, 37}

```

[illegible]

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(*Case2: the streaming corruption is not corrected,
the damping is protected, Lambda is too large*)
(*Combine residual across all excitation frequencies*)
Residual02 = Residual /. Thread[Table[ $\delta A_{s,0}[j2]$ , {j2, 1, Length[OmegaList]}]  $\rightarrow$  0];
(*Merge physical parameters and noise
correction coefficients into full variable set*)
Variable02 = Join[Var,
Table[Join[CorrectCoeffjVar[j2]], {j2, 1, Length[OmegaList]}] // Flatten];
(*Compute Jacobian matrix of residual with respect to unknown variables*)
JacobiFun02 = D[Residual02, {Variable02, 1}];
(*Initial guess of unknown parameter vector*)
InitGuess02 = ReplacePart[ReplacePart[0.2 * Variable02 /. Par01,
Thread[Range[Length[Var] + 1, Length[Variable02]]  $\rightarrow$  0]], Thread[{6, 8, 9}  $\rightarrow$  1]];
(*Indices of physical terms forced to be retained during thresholding*)
InitbigIndices02 = Join[{2}, Range[Length[Var] + 1, Length[Variable02]]];
```

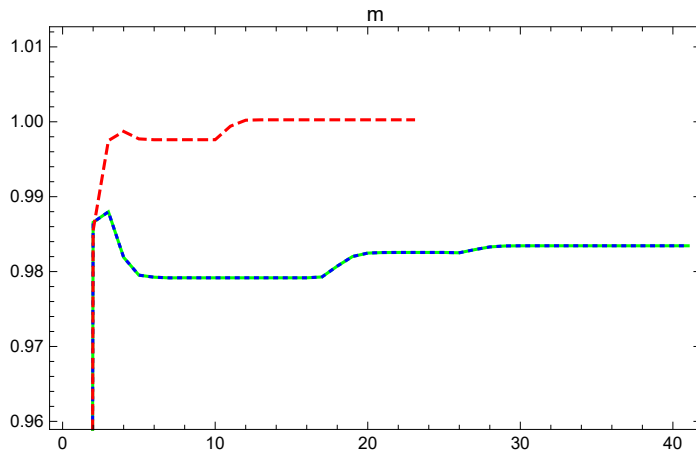
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(*Execute proposed LM-STLS nonlinear sparse regression*)
Xi02 = NonLinearSparsifyDynamics[JacobiFun02, Residual02, Variable02,
    InitGuess02, ConvTol, maxLMIter, Lambda, maxSparseIter, InitbigIndices02];
{1, 39}
{2, 38}
{3, 38}
{4, 38}
{5, 38}
{6, 38}
{7, 38}
{8, 38}
{9, 38}
{10, 38}
```

[illegible]

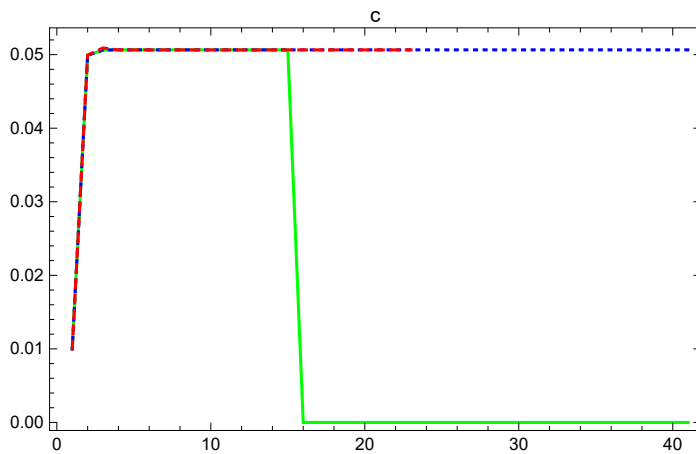
```
(*Case3: the streaming corruption is corrected,
the damping is protected, Lambda is too large*)
(*Combine residual across all excitation frequencies*)
Residual03 = Residual;
(*Merge physical parameters and noise
correction coefficients into full variable set*)
Variable03 = Join[Var, Table[Join[CorrectCoeffjVar[j2], { $\delta A_{s,0}[j2]$ }],
{j2, 1, Length[Omegalist]}] // Flatten];
(*Compute Jacobian matrix of residual with respect to unknown variables*)
JacobiFun03 = D[Residual03, {Variable03, 1}];
(*Initial guess of unknown parameter vector*)
InitGuess03 = ReplacePart[ReplacePart[0.2 * Variable03 /. Par01,
Thread[Range[Length[Var] + 1, Length[Variable03]]  $\rightarrow$  0]], Thread[{6, 8, 9}  $\rightarrow$  1]];
(*Indices of physical terms forced to be retained during thresholding*)
InitbigIndices03 = Join[{2}, Range[Length[Var] + 1, Length[Variable03]]];
```



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(*Plot Fig5*)
Fig5a1 = ListLinePlot[Table[Xi01[[1]][[j1]][[1]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5a2 = ListLinePlot[Table[Xi02[[1]][[j1]][[1]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5a3 = ListLinePlot[Table[Xi03[[1]][[j1]][[1]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5a = Show[Fig5a1, Fig5a2, Fig5a3, PlotRange → {0.96, 1.01}, PlotLabel → "m"]
```



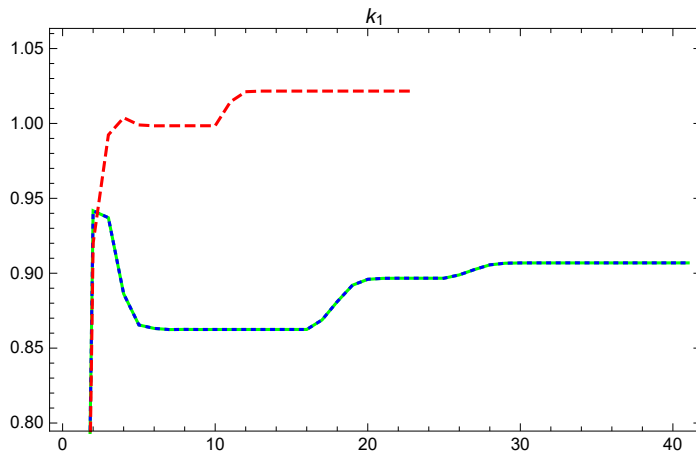
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Fig5b1 = ListLinePlot[Table[Xi01[[1]][[j1]][[2]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5b2 = ListLinePlot[Table[Xi02[[1]][[j1]][[2]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5b3 = ListLinePlot[Table[Xi03[[1]][[j1]][[2]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5b = Show[Fig5b1, Fig5b2, Fig5b3, PlotRange → All, PlotLabel → "c"]
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Fig5c1 = ListLinePlot[Table[Xi01[[1]][[j1]][[3]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5c2 = ListLinePlot[Table[Xi02[[1]][[j1]][[3]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5c3 = ListLinePlot[Table[Xi03[[1]][[j1]][[3]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5c = Show[Fig5c1, Fig5c2, Fig5c3, PlotRange → {0.80, 1.05}, PlotLabel → "k1"]

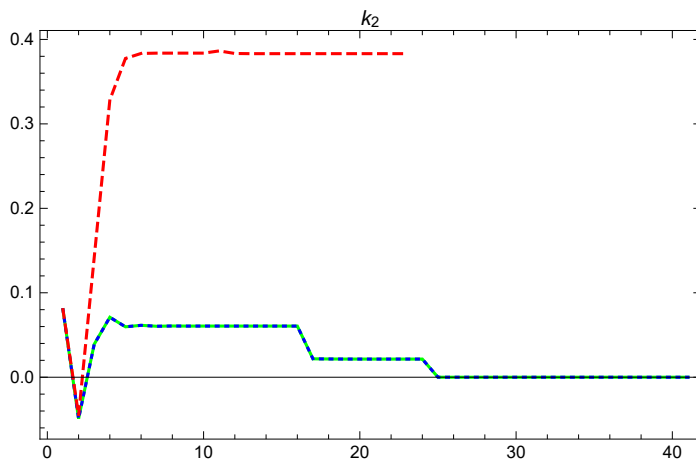
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Fig5d1 = ListLinePlot[Table[Xi01[[1]][[j1]][[4]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5d2 = ListLinePlot[Table[Xi02[[1]][[j1]][[4]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5d3 = ListLinePlot[Table[Xi03[[1]][[j1]][[4]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5d = Show[Fig5d1, Fig5d2, Fig5d3, PlotRange → All, PlotLabel → "k2"]

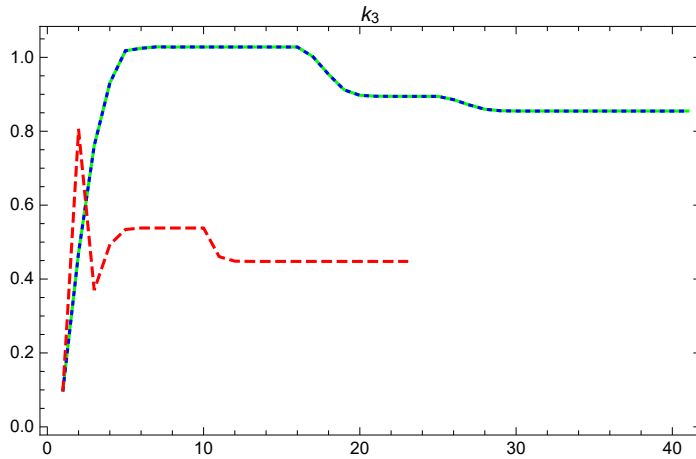
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Fig5e1 = ListLinePlot[Table[Xi01[[1]][[j1]][[5]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5e2 = ListLinePlot[Table[Xi02[[1]][[j1]][[5]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5e3 = ListLinePlot[Table[Xi03[[1]][[j1]][[5]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5e = Show[Fig5e1, Fig5e2, Fig5e3, PlotRange → All, PlotLabel → "k3"]

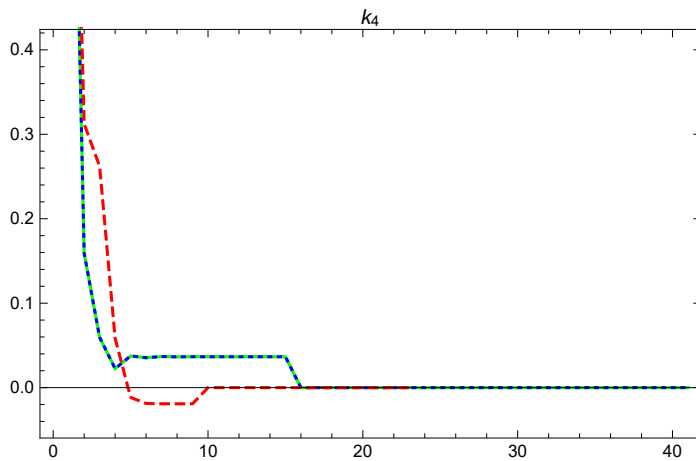
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Fig5f1 = ListLinePlot[Table[Xi01[[1]][[j1]][[6]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5f2 = ListLinePlot[Table[Xi02[[1]][[j1]][[6]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5f3 = ListLinePlot[Table[Xi03[[1]][[j1]][[6]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5f = Show[Fig5f1, Fig5f2, Fig5f3, PlotRange → {-0.05, 0.4}, PlotLabel → "k4"]

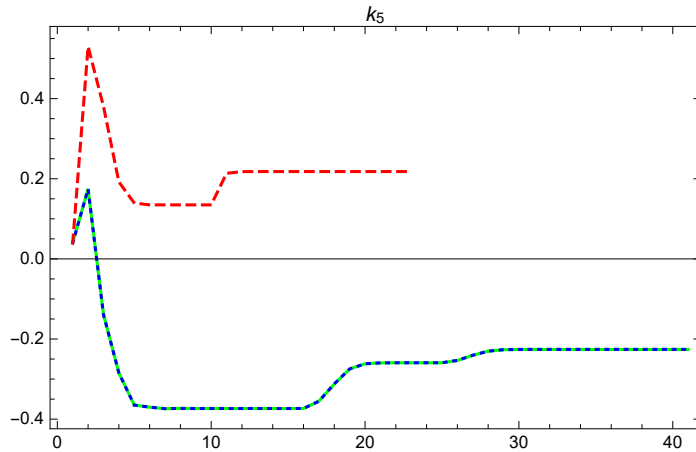
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Fig5g1 = ListLinePlot[Table[Xi01[[1]][[j1]][[7]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5g2 = ListLinePlot[Table[Xi02[[1]][[j1]][[7]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5g3 = ListLinePlot[Table[Xi03[[1]][[j1]][[7]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5g = Show[Fig5g1, Fig5g2, Fig5g3, PlotRange → All, PlotLabel → "k5"]

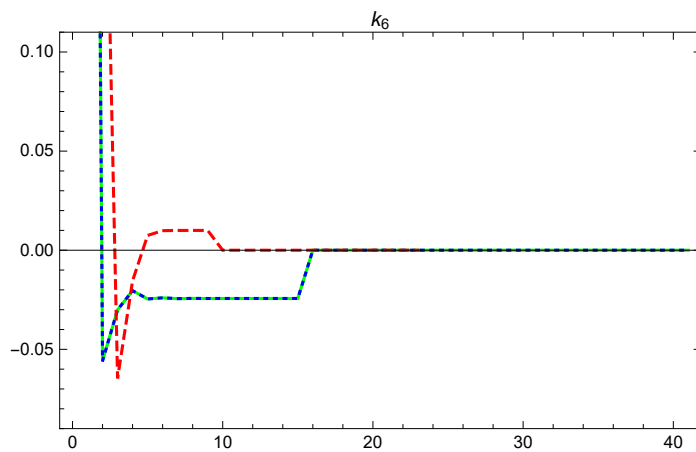
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Fig5h1 = ListLinePlot[Table[Xi01[[1]][[j1]][[8]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5h2 = ListLinePlot[Table[Xi02[[1]][[j1]][[8]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5h3 = ListLinePlot[Table[Xi03[[1]][[j1]][[8]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5h = Show[Fig5h1, Fig5h2, Fig5h3, PlotRange → {-0.08, 0.1}, PlotLabel → "k6"]

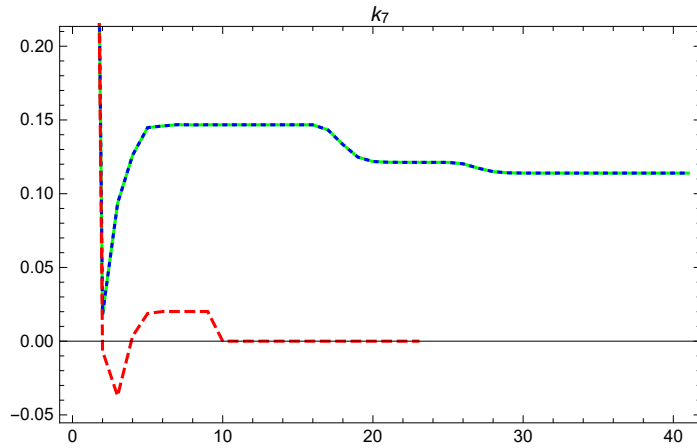
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Fig5i1 = ListLinePlot[Table[Xi01[[1]][[j1]][[9]], {j1, 1, Length[Xi01[[1]]}],
  PlotRange → All, PlotStyle → {Green}, Frame → True];
Fig5i2 = ListLinePlot[Table[Xi02[[1]][[j1]][[9]], {j1, 1, Length[Xi02[[1]]}],
  PlotRange → All, PlotStyle → {Blue, Dotted}, Frame → True];
Fig5i3 = ListLinePlot[Table[Xi03[[1]][[j1]][[9]], {j1, 1, Length[Xi03[[1]]}],
  PlotRange → All, PlotStyle → {Red, Dashed}, Frame → True];
Fig5i = Show[Fig5i1, Fig5i2, Fig5i3, PlotRange → {-0.05, 0.20}, PlotLabel → "k7"]

```



```
Fig5 = {Fig5a, Fig5b, Fig5c, Fig5d, Fig5e, Fig5f, Fig5g, Fig5h, Fig5i}
```

